

Laxmi Charitable Trust's
Sheth L.U.J College of Arts & Sir M.V. College of Science and Commerce
Department of Information Technology (B.Sc.I.T Semester V)
Full Stack development MERN Practical

Practical – II

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Class: TYIT	Batch: 1
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AIM: Core JavaScript Concepts for MERN

A. Create functions and call them with parameters.

CODE A:

```
function calculateSalary(name , basicSalary){
    let bonus = 5000;
    let totalSalary = basicSalary + bonus;

    console.log("===== Employee Details =====");
    console.log("Employee Name: ",name);
    console.log("Basic Salary: ",basicSalary);
    console.log("Bonus: ",bonus);
    console.log("Total Salary: ",totalSalary);
}
//Calling Function
calculateSalary("Himani" , 200000);
```

OUTPUT:

```
himanimalankar@Himanis-Macbook Practical_2 % node Practical_2A.js
===== Employee Details =====
Employee Name:  Himani
Basic Salary:  200000
Bonus:  5000
Total Salary:  205000
```

CODE B:

```
function BillGeneration(customerName , itemName , quantity , pricePerItem){
    let amount = quantity * pricePerItem;
    let gst = amount * 0.18;
    let totalAmount = amount + gst;

    console.log("===== Your Bill =====");
    console.log("Customer Name:",customerName);
    console.log("Item Name:",itemName);
    console.log("Quantity:",quantity);
    console.log("Price/Item: ₹",pricePerItem);
    console.log("Amount: ₹",amount);
```

```
console.log("GST (18%): ₹",gst);
console.log("Total Bill: ₹",totalAmount);
}

//Calling Function
BillGeneration("Himani Malankar", "MacBook", 1, 245000);
```

OUTPUT:

```
● himanimalankar@Himanis-Macbook Practical_2 % node Practical_2A.js
==== Your Bill ====
Customer Name: Himani Malankar
Item Name: MacBook
Quantity: 1
Price/Item: ₹ 245000
Amount: ₹ 245000
GST (18%): ₹ 44100
Total Bill: ₹ 289100
```

B. Work with arrays and objects in JavaScript.

CODE:

```
//array
let fruits = ['Mango' , 'Lichi' , 'Dragonfruit' , 'Watermelon' , 'Mangosteem'];

//objects
let student = {
  name : 'Himani Malankar',
  age : 20,
  city : "Mumbai"
};
console.log("Fruits: ",fruits[2]);
console.log("4th fruit: ",fruits[4]);

console.log("Student Name: ",student.name);
console.log("Student Age: ",student.age);
console.log("City where student live: ",student.city);
```

OUTPUT:

```
● himanimalankar@Himanis-Macbook Practical_2 % node Practical_2B.js
Fruits:  Dragonfruit
4th fruit:  Mangosteem
Student Name:  Himani Malankar
Student Age:  20
City where student live:  Mumbai
```

C. Demonstrate use of arrow functions.

CODE:

```
//Arrow symbol
let add = (a , b) =>{
    return a + b;
};

let sub = (a , b) => {
    return a - b;
};

let multiply = (a , b) => {
    return a * b;
};

let div = (a , b) => {
    return a / b;
}

console.log("Addition: ",add(10 , 20));
console.log("Subtraction: ",sub(20 , 10));
console.log("Multiplication: ",multiply(5 , 6));
console.log("Division: ",div(10 , 2));
```

OUTPUT:

```
● himanimalankar@Himanis-Macbook Practical_2 % node Practical_2C.js
Addition:  30
Subtraction:  10
Multiplication:  30
Division:  5
```

D. Write a program using array methods (map, filter, reduce).

map: Perform one operation on every array.

filter: sets a condition for that particular condition.

reduce: prints single value (aggregate value).

CODE:

```
let numbers = [10 , 20 , 30 , 40 , 50];
//map()
let doubled = numbers.map(num => num * 2);

//filter()
let greater25 = numbers.filter(num => num > 25);

//reduce()
let total = numbers.reduce((sum , num) => sum + num ,0);

console.log("Original Array: ",numbers);
console.log("Map: ",doubled);
console.log("Filter: ",greater25);
console.log("Reduce: ",total);
```

OUTPUT:

```
● himanimalankar@Himanis-Macbook Practical_2 % node Practical_2D.js
Original Array: [ 10, 20, 30, 40, 50 ]
Map: [ 20, 40, 60, 80, 100 ]
Filter: [ 30, 40, 50 ]
Reduce: 150
```

E. Create a program that manipulates objects and displays output.

CODE:

```
// Creating object
let student = {
  rollNo : "T016",
  name : "Himani Malankar",
  marks : 90
}

console.log("Original Object");
console.log(student);

//Accessing object Properties
console.log("\nAccessing object Properties");
console.log("Name:",student.name);
console.log("Marks:",student.marks);

//Updating Properties
student.marks = 95;
console.log("\nAfter Updating Marks");
console.log(student);

//Adding new property
student.city = "Mumbai";
console.log("\nAfter adding new property");
console.log(student);

//Delete a property
delete student.rollNo
console.log("\nAfter deleteing one property");
console.log(student)
```

OUTPUT:

```
himanimalankar@Himanis-Macbook Practical_2 % node Practical_2E.js
Original Object
{ rollNo: 'T016', name: 'Himani Malankar', marks: 90 }

Accessing object Properties
Name: Himani Malankar
Marks: 90

After Updating Marks
{ rollNo: 'T016', name: 'Himani Malankar', marks: 95 }

After adding new property
{ rollNo: 'T016', name: 'Himani Malankar', marks: 95, city: 'Mumbai' }

After deleteing one property
{ name: 'Himani Malankar', marks: 95, city: 'Mumbai' }
```

F. Implement a small JavaScript program combining functions, arrays, and objects.

CODE:

```
//Array of objects
let student = [
  {
    rollNo : "T005",
    name : "Ritesh",
    marks : 85
  },
  {
    rollNo : "T014",
    name : "Saqlain",
    marks : 90
  },
  {
    rollNo : "T016",
    name : "Himani",
    marks : 90
  }
];

//function display
function display(s)
{
  console.log("===== STUDENT DETAILS =====");
  console.log("-----");
  for (let stud of s)
  {
    console.log(`Roll No : ${stud.rollNo}`);
    console.log(`Name : ${stud.name}`);
    console.log(`Marks : ${stud.marks}`);
    console.log("-----");
  }
}

//Function Call
display(student);
```

OUTPUT:

```
● himanimalankar@Himanis-Macbook Practical_2 % node Practical_2F.js
===== STUDENT DETAILS =====
-----
Roll No : T005
Name : Ritesh
Marks : 85
-----
Roll No : T014
Name : Saqlain
Marks : 90
-----
Roll No : T016
Name : Himani
Marks : 90
```